

TikTok Video View Summary

Setup & Data Import

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v forcats   1.0.1      v readr      2.2.0
```

```
v ggplot2   4.0.3      v stringr    1.6.0
```

```
v lubridate 1.9.5      v tibble     3.3.1
```

```
v purrr     1.2.2      v tidyr      1.3.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
views <- read.csv("data/raw/video_view.csv")
```

Data Inspection

```
count_rows <- nrow(views)
print(count_rows)
```

```
[1] 15000
```

```
str(views)
```

```
'data.frame':  15000 obs. of  16 variables:
 $ video_id          : int  1 2 3 4 5 6 7 8 9 10 ...
 $ creator_id        : int  66 62 176 22 39 148 158 30 200 19 ...
 $ video_length_sec   : int  14 47 37 29 28 36 30 34 25 48 ...
 $ published_at       : int  1767297237 1768475971 1766815457 1755689136 1762696418 ...
 $ video_url          : chr  "https://www.tiktok.com/@olivialive2824/video/9000000000000000000"
 $ impressions_n      : int  4 0 14 67 25 14 23 19 6 86 ...
 $ watched_n         : int  3 0 10 49 18 7 19 14 4 64 ...
 $ watch_rate         : num  0.75 NA 0.714 0.731 0.72 ...
 $ avg_watch_seconds_when_watched: num  10.7 NA 18.6 15.4 14.4 ...
 $ total_watch_seconds : int  32 0 186 754 260 133 322 273 73 1517 ...
 $ avg_watch_share     : num  0.571 NA 0.359 0.388 0.371 ...
 $ distinct_watchers_n : int  3 0 10 43 15 7 19 14 4 59 ...
 $ first_impression_at : int  1768287244 NA 1766868873 1755661745 1762792600 1765000000 ...
 $ last_impression_at  : int  1769439546 NA 1769741156 1769429852 1769860569 1769100000 ...
 $ first_watch_at      : int  1768287244 NA 1766868873 1755661745 1762792600 1765000000 ...
 $ last_watch_at       : int  1769439546 NA 1769741156 1769429852 1769860569 1769100000 ...
```

```
head(views)
```

	video_id	creator_id	video_length_sec	published_at
1	1	66	14	1767297237
2	2	62	47	1768475971
3	3	176	37	1766815457
4	4	22	29	1755689136
5	5	39	28	1762696418

```

6           6           148           36   1764970157
video_url
1 https://www.tiktok.com/@olivialive2824/video/9000000000000000001
2 https://www.tiktok.com/@isabellabrown/video/9000000000000000002
3 https://www.tiktok.com/@isabellawalker/video/9000000000000000003
4 https://www.tiktok.com/@lucastaylor/video/9000000000000000004
5 https://www.tiktok.com/@avabrown/video/9000000000000000005
6 https://www.tiktok.com/@harperwalker_1/video/9000000000000000006
  impressions_n watched_n watch_rate avg_watch_seconds_when_watched
1           4           3  0.7500000           10.66667
2           0           0           NA           NA
3          14          10  0.7142857           18.60000
4          67          49  0.7313433           15.38776
5          25          18  0.7200000           14.44444
6          14           7  0.5000000           19.00000
total_watch_seconds avg_watch_share distinct_watchers_n first_impression_at
1           32           0.5714286           3           1768287244
2           0           NA           0           NA
3          186           0.3590734          10           1766868873
4          754           0.3880597          43           1755661745
5          260           0.3714286          15           1762792600
6          133           0.2638889           7           1765069292
last_impression_at first_watch_at last_watch_at
1      1769439546      1768287244      1769439546
2           NA           NA           NA
3      1769741156      1766868873      1769741156
4      1769429852      1755661745      1769429852
5      1769860569      1762792600      1769860569
6      1769101217      1765069292      1769101217

```

```
summary(views)
```

```

video_id      creator_id  video_length_sec  published_at
Min.   :    1  Min.   : 1.00  Min.   :10.00  Min.   :1.754e+09
1st Qu.: 3751  1st Qu.: 48.00  1st Qu.:23.00  1st Qu.:1.758e+09
Median : 7500  Median : 97.00  Median :30.00  Median :1.762e+09
Mean   : 7500  Mean   : 98.06  Mean   :30.08  Mean   :1.762e+09
3rd Qu.:11250  3rd Qu.:146.00  3rd Qu.:37.00  3rd Qu.:1.766e+09
Max.   :15000  Max.   :200.00  Max.   :60.00  Max.   :1.770e+09

```

```

video_url      impressions_n      watched_n      watch_rate
Length   :15000  Min.   : 0.00  Min.   : 0.00  Min.   :0.0000

```

N.unique :15000	1st Qu.: 9.00	1st Qu.: 6.00	1st Qu.:0.6250
N.blank : 0	Median : 22.00	Median : 15.00	Median :0.6957
Min.nchar: 58	Mean : 29.76	Mean : 20.45	Mean :0.6947
Max.nchar: 70	3rd Qu.: 41.00	3rd Qu.: 28.00	3rd Qu.:0.7679
	Max. :230.00	Max. :158.00	Max. :1.0000
			NAs :384

avg_watch_seconds_when_watched	total_watch_seconds	avg_watch_share
Min. : 1.00	Min. : 0.0	Min. :0.0000
1st Qu.:11.75	1st Qu.: 92.0	1st Qu.:0.3148
Median :16.28	Median : 226.0	Median :0.3747
Mean :16.42	Mean : 336.5	Mean :0.3794
3rd Qu.:20.56	3rd Qu.: 464.0	3rd Qu.:0.4352
Max. :52.00	Max. :3945.0	Max. :1.0000
NAs :536		NAs :384

distinct_watchers_n	first_impression_at	last_impression_at
Min. : 0.00	Min. :1.754e+09	Min. :1.754e+09
1st Qu.: 6.00	1st Qu.:1.758e+09	1st Qu.:1.769e+09
Median : 15.00	Median :1.762e+09	Median :1.770e+09
Mean : 19.35	Mean :1.762e+09	Mean :1.769e+09
3rd Qu.: 27.00	3rd Qu.:1.766e+09	3rd Qu.:1.770e+09
Max. :135.00	Max. :1.770e+09	Max. :1.770e+09
	NAs :384	NAs :384

first_watch_at	last_watch_at
Min. :1.754e+09	Min. :1.754e+09
1st Qu.:1.758e+09	1st Qu.:1.769e+09
Median :1.762e+09	Median :1.770e+09
Mean :1.762e+09	Mean :1.769e+09
3rd Qu.:1.766e+09	3rd Qu.:1.770e+09
Max. :1.770e+09	Max. :1.770e+09
NAs :384	NAs :384

```
colSums(is.na(views))
```

video_id	creator_id
0	0
video_length_sec	published_at
0	0
video_url	impressions_n
0	0
watched_n	watch_rate
0	384
avg_watch_seconds_when_watched	total_watch_seconds

	536		0
avg_watch_share		distinct_watchers_n	
	384		0
first_impression_at		last_impression_at	
	384		384
first_watch_at		last_watch_at	
	384		384

```
names(views)
```

```
[1] "video_id"           "creator_id"
[3] "video_length_sec"   "published_at"
[5] "video_url"          "impressions_n"
[7] "watched_n"          "watch_rate"
[9] "avg_watch_seconds_when_watched" "total_watch_seconds"
[11] "avg_watch_share"     "distinct_watchers_n"
[13] "first_impression_at" "last_impression_at"
[15] "first_watch_at"      "last_watch_at"
```

```
glimpse(views)
```

```
Rows: 15,000
```

```
Columns: 16
```

```
$ video_id           <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, ~
$ creator_id         <int> 66, 62, 176, 22, 39, 148, 158, 30, 200, ~
$ video_length_sec   <int> 14, 47, 37, 29, 28, 36, 30, 34, 25, 48, ~
$ published_at       <int> 1767297237, 1768475971, 1766815457, 175~
$ video_url          <chr> "https://www.tiktok.com/@olivialive2824~
$ impressions_n      <int> 4, 0, 14, 67, 25, 14, 23, 19, 6, 86, 4, ~
$ watched_n         <int> 3, 0, 10, 49, 18, 7, 19, 14, 4, 64, 3, ~
$ watch_rate         <dbl> 0.7500000, NA, 0.7142857, 0.7313433, 0.~
$ avg_watch_seconds_when_watched <dbl> 10.66667, NA, 18.60000, 15.38776, 14.44~
$ total_watch_seconds <int> 32, 0, 186, 754, 260, 133, 322, 273, 73~
$ avg_watch_share    <dbl> 0.5714286, NA, 0.3590734, 0.3880597, 0.~
$ distinct_watchers_n <int> 3, 0, 10, 43, 15, 7, 19, 14, 4, 59, 3, ~
$ first_impression_at <int> 1768287244, NA, 1766868873, 1755661745, ~
$ last_impression_at <int> 1769439546, NA, 1769741156, 1769429852, ~
$ first_watch_at     <int> 1768287244, NA, 1766868873, 1755661745, ~
$ last_watch_at      <int> 1769439546, NA, 1769741156, 1769429852, ~
```

General Summary

```
overview <- views %>%
  summarize(
    videos = n(),
    creators = n_distinct(creator_id),
    avg_length = mean(video_length_sec),
    avg_impressions = mean(impressions_n),
    avg_watched = mean(watched_n),
    avg_watch_rate = mean(watch_rate, na.rm=TRUE),
    avg_watch_share = mean(avg_watch_share, na.rm=TRUE),
    avg_watch_seconds = mean(avg_watch_seconds_when_watched, na.rm=TRUE)
  )

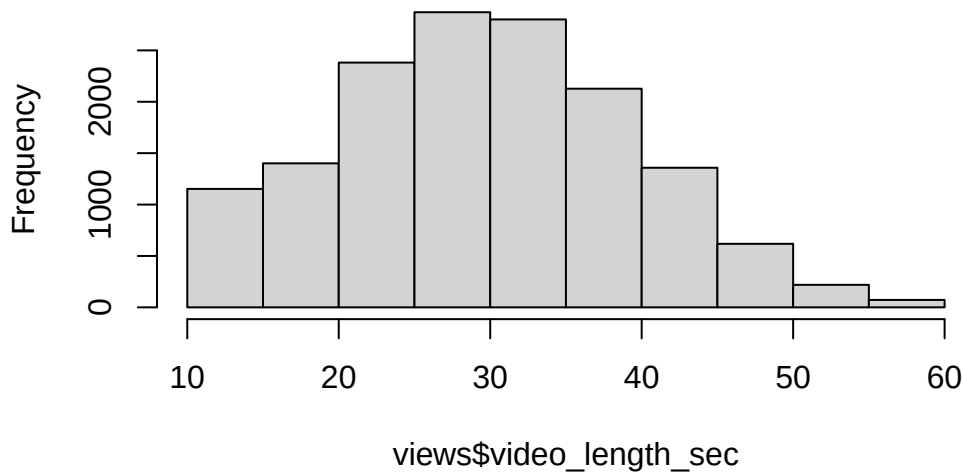
print(overview)
```

	videos	creators	avg_length	avg_impressions	avg_watched	avg_watch_rate
1	15000	200	30.08247	29.76147	20.4512	0.6946644
	avg_watch_share	avg_watch_seconds				
1	0.3794281	16.42387				

```
views$watched_pct <- views$watched_n / views$impressions_n * 100

hist(views$video_length_sec)
```

Histogram of views\$video_length_sec



```
goodvideos <- subset(views, watch_rate > 0.8)
nrow(goodvideos)
```

```
[1] 2458
```

Top Videos

Top Video by Impressions

```
topvideo <- views[views$impressions_n == max(views$impressions_n), ]
print(topvideo)
```

```
video_id creator_id video_length_sec published_at
13801    13801      197                25  1754027509
video_url
13801 https://www.tiktok.com/@elijahofficial7391/video/90000000000000013801
impressions_n watched_n watch_rate avg_watch_seconds_when_watched
13801      230      156  0.6782609                14.29487
total_watch_seconds avg_watch_share distinct_watchers_n
```

13801	2230	0.3878261	134
	first_impression_at	last_impression_at	first_watch_at last_watch_at
13801	1754008389	1769541954	1754008389 1769541954
	watched_pct		
13801	67.82609		

Top Videos by Watch Rate

```
top_watch <- views %>%
  select(video_id, creator_id, watch_rate, impressions_n) %>%
  filter(impressions_n >= 60) %>%
  arrange(desc(watch_rate))

head(top_watch)
```

	video_id	creator_id	watch_rate	impressions_n
1	3683	16	0.8833333	60
2	2418	125	0.8548387	62
3	7351	199	0.8536585	82
4	11903	176	0.8382353	68
5	5584	97	0.8311688	77
6	10372	174	0.8279570	93

Creator Performance

```
creator_summary <- views %>%
  group_by(creator_id) %>%
  summarise(
    videos_n = n(),
    impressions_total = sum(impressions_n, na.rm = TRUE),
    watch_rate_avg = mean(watch_rate, na.rm = TRUE)
  ) %>%
  arrange(desc(impressions_total))

head(creator_summary)
```

A tibble: 6 x 4

	creator_id	videos_n	impressions_total	watch_rate_avg
	<int>	<int>	<int>	<dbl>
1	7	152	5791	0.700
2	96	184	5734	0.665
3	49	173	5430	0.719
4	196	143	5152	0.680
5	134	163	5105	0.696
6	75	165	4796	0.704

```
creator_engagement <- views %>%
  mutate(engagement = watched_n / impressions_n) %>%
  group_by(creator_id) %>%
  summarise(mean_engagement = mean(engagement, na.rm = TRUE)) %>%
  arrange(desc(mean_engagement))

head(creator_engagement)
```

```
# A tibble: 6 x 2
  creator_id mean_engagement
    <int>         <dbl>
1     117         0.749
2     143         0.743
3     154         0.741
4      55         0.738
5     114         0.736
6     184         0.732
```

Regression Analysis

This analysis investigates the relationship between `score_satiation_penalty` and `feed_rank`.

```
impressions <- read_csv("data/raw/impressions_clean.csv")
```

```
Rows: 94790 Columns: 14
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr   (3): source_bucket, mission_ids, creator_display_name
```

```
dbl  (10): impression_id, session_id, user_id, video_id, creator_id, feed_ra...
```

```
dtm   (1): shown_at
```

- i Use ``spec()`` to retrieve the full column specification for this data.
- i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
modell1 <- lm(  
  feed_rank ~ score_satiation_penalty,  
  data = impressions  
)  
  
summary(modell1)
```

Call:

```
lm(formula = feed_rank ~ score_satiation_penalty, data = impressions)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.8631	-2.1181	-0.8676	1.2645	28.1766

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.61762	0.01901	137.67	<2e-16 ***
score_satiation_penalty	4.08746	0.05350	76.41	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.188 on 94788 degrees of freedom

Multiple R-squared: 0.05802, Adjusted R-squared: 0.05801

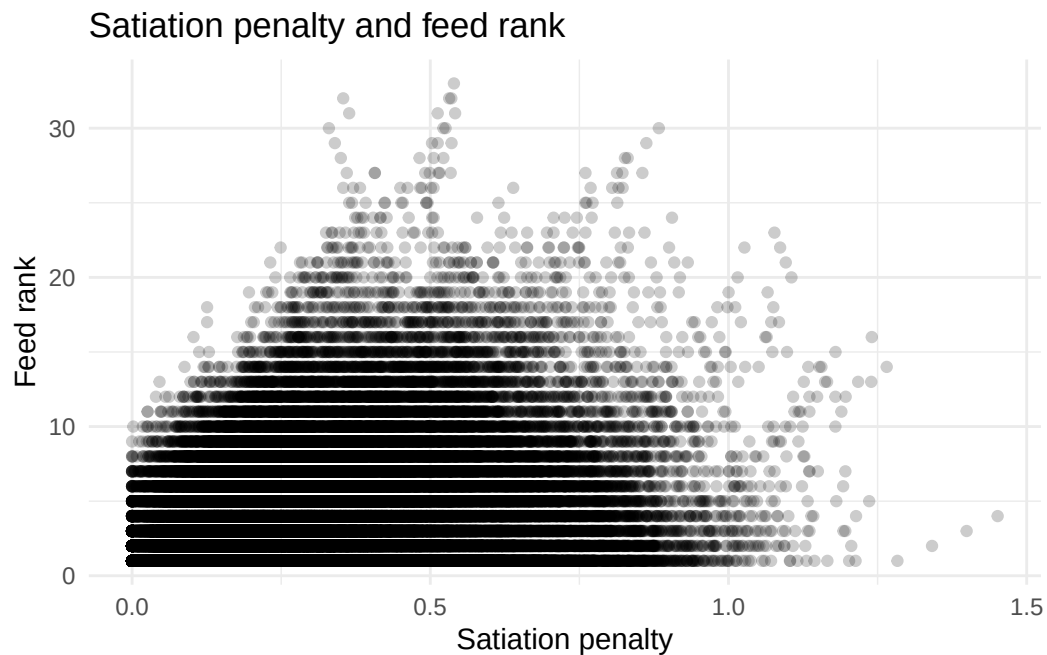
F-statistic: 5838 on 1 and 94788 DF, p-value: < 2.2e-16

The first model shows a significant positive relationship between `score_satiation_penalty` and `feed_rank`. A one-unit increase in `score_satiation_penalty` is associated with an increase of approximately 4.09 in `feed_rank`. The model explains approximately 5.8% of the variation in `feed_rank`.

Relationship Plot

```
ggplot(  
  impressions,  
  aes(x = score_satiation_penalty, y = feed_rank)
```

```
) +
  geom_point(alpha = 0.2) +
  labs(
    title = "Satiation penalty and feed rank",
    x = "Satiation penalty",
    y = "Feed rank"
  ) +
  theme_minimal()
```



The plot shows the positive relationship between `score_satiation_penalty` and `feed_rank` found in the first regression model.

Extended Regression Model

```
model2 <- lm(
  feed_rank ~ score_satiation_penalty + score_category_match + score_creator_match,
  data = impressions
)

summary(model2)
```

```
Call:
lm(formula = feed_rank ~ score_satiation_penalty + score_category_match +
    score_creator_match, data = impressions)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-6.8320	-2.1129	-0.8704	1.2678	28.4427

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.60612	0.01935	134.710	< 2e-16 ***
score_satiation_penalty	4.12848	0.05477	75.382	< 2e-16 ***
score_category_match	0.09961	0.01261	7.901	2.81e-15 ***
score_creator_match	-1.83569	0.21155	-8.677	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.186 on 94786 degrees of freedom

Multiple R-squared: 0.05926, Adjusted R-squared: 0.05923

F-statistic: 1990 on 3 and 94786 DF, p-value: < 2.2e-16

After adding `score_category_match` and `score_creator_match`, the relationship between `score_satiation_penalty` and `feed_rank` remains similar. The extended model explains approximately 5.9% of the variation in `feed_rank`.